HAO LIU

hl4220@nyu.edu | (+1) 5512351734

EDUCATION

University of California, Irvine (UCI) Starting Sep 2026

Cognitive Sciences Ph.D. with concentration (M.S.) in Cognitive Neuroscience

- Doctoral advisor: Mark Steyvers, Aaron Bornstein (co-supervise)

New York University (NYU) Sep 2024– May 2026

Master of Science (Cognitive Psychology)

- Current Major GPA: 3.76 /4.0

Tsinghua University (THU) June 2021– June 2024

Bachelor of Engineering (Computer Science and Technology)

- Computer Science Major GPA: 3.23 /4.0
- Core Courses: Fundamentals of Programming: 4.0
  Discrete Mathematics:3.6
  Linear Algebra: 3.6
  Foundation of Object-Oriented Programming: 4.0
  Artificial Neural Networks: 4.0
  Advances in Autonomous Driving and Intelligent Vehicles: 4.0 (A+)
  English Reading and Writing for academic purpose: 4.0 (A)
  English listening and speaking for academic purpose: 4.0 (A+)

Tsinghua University (THU) Sep 2020– June 2021

Electronic Engineering

Beijing Institute for General Artificial Intelligence (BIGAI) Sep 2022– June 2024

Intern (Multiagent Group)

Massachusetts Institute of Technology (MIT) June 2023– September 2023

Summer Internship (Department of Brain and Cognitive Sciences)

PUBLICATIONS

Huang, Y., Wang, X., **Liu, H.**, Kong, F., Qin, A., Tang, M., Zhu, S., Bi, M., Qi, S., & Feng, X. (2024). AdaSociety: An adaptive environment with social structures for multi-agent decision-making. *Advances in Neural Information Processing Systems, 37* (Datasets and Benchmarks Track).

Liu, H., Mattar, M., Michelmann, S. (in prep.). Learning distributed representations with complementary components.

Yu, J., Wu, X., **Liu, H.**, Guo, W., & Xing, X. (2025). BlockScan: Detecting Anomalies in Blockchain Transactions. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

CONFERENCE CONTRIBUTIONS

NeurIPS 2025 Poster: BlockScan: Detecting Anomalies in Blockchain Transactions.
(https://arxiv.org/pdf/2410.04039)

Advisor: Xinyu Xing, Wenbo Guo / Northwestern (Remote) March 2024– June 2024

- Help build and test LLM based anomaly detection algorithm on blockchain transaction.
- Carefully clean and analyze the transaction data, including developing multi-thread algorithms to efficiently preprocess the data.
- Train and test different LLM models on fitting the dataset, LoRA for example.

NeurIPS 2024 Benchmark Track, Poster: AdaSociety: An Adaptive Environment with Social Structures for Multi-Agent Decision-Making
(https://neurips.cc/virtual/2024/poster/97511)

Advisor: Siyuan Qi, Xue Feng, Songchun Zhu / Peking University June 2023– June 2024

- Built a Grid world Based environment benchmark for studying generalized social behavior. The main dynamic of the environment consists of 3 parts: resource, event and agent. Agents could gather resources to make higher products through engaging events.
- Using RLLib to deploy classic algorithms like MAPPO, MADDPG to see the result.

AWARDS

YINGCAIJIHUA (Research training for science and technology innovation for high school students) October 2019

- Outstanding paper award
- Outstanding Student award

RESEARCH EXPERIENCE

Main MA project: Learning distributed representations with complementary components

Advisor: Sebastian Michelmann, Marcelo Matter/ NYU October 2024– Now

- We simulate distributed memory representations with a generative neural network model (Stable Diffusion). These representations are learned with various levels of encoding strength that result in more accurate memory representations.
- Neo-cortical statistical knowledge is modeled by the weights of the network and hippocampus is implemented as a Modern Hopfield Network that stores the starting points to re-generate distributed representations from a low-dimensional latent.
- The model can perform the Mnemonic Similarity Task (MST) with various depth of encoding. Its output patterns mirror findings from memory impairment in human behavior.
- A fine-tuned "expert" models can learn new information more effectively if the stimulus material pertains to the area of expertise.
- I related the representational similarity between images in our model's latent representations to the representational similarity in the Natural Scenes Dataset (NSD). Deeper encoding in our model yields better representational alignment with regions the medial temporal lobe (MTL)
- I tested model predictions by designing a human behavioral experiment in which participants reconstruct masked images in a modified mnemonic similarity task.

Summer research project: Watch-Talk-Help – Multiagent communication based on Large Language Model and Theory of Mind

Advisor: Tianmin Shu, Joshua Tenenbaum / MIT June 2023– September 2023

- My project aimed to determine when to engage in natural language communication to bring the goals between humans and robots into alignment and to effectively plan together.
- I improved the model underlying the previous articles "watch-and-help" (Puig, X., et al., 2020) and "online-watch-and-help" (Puig, X., et al., 2023) by adding natural language communication parsed using LLM based agents' believes.
- Based on this natural language interaction, I defined rules for updating the agent's belief and goals.

OTHER RESEARCH PROJECTS

Pairwise learning / Functional embedding

Advisor: Zhiting Hu / UCSD(Remote) June 2022– September 2022

- Developed a unified framework of data & parameter space by learning pairs of data (leveraging CNNs & Transformers)

Stock prediction based on large scale pretrained models February 2022–September 2022

Advisor: Maosong Sun / Tsinghua University

- Built a transformer-based Chinese financial model by crawling 15GB of market text and designing a fine-tuning pipeline across Macro, Meso, and Micro levels.

A Self-supervised pretraining model based on Encoder-Decoder-Masker architecture December 2021–February 2022

Advisor: None

- Built a self-supervised encoder–decoder–unmasker on ImageNet-1K (hand-implemented ViT backbone), achieving competitive transfer to classification and low-level vision tasks.

A Q-A model based on query mechanism September 2021–November 2022

Advisor: Minlie Huang / Tsinghua University

- Curated 1M+ Zhihu QA pairs, built two different query mechanisms (BM25 and BERT-based relevance), integrated retrieval into LongLM pretraining, and surpassed baselines on BLEU/ROUGE.

Pollution Image classification system December 2017–October 2019(High school)

Advisor: Mingming Cheng / Nankai University

- Crawled 10k+ pollution images and trained an SVM classifier using hand-crafted features (dark channel, information entropy, and HSV color histograms).

Additional open coursework

- Open course: Stanford CS231n, Computer Vision September 2020–December 2020
- Open course: Stanford CS224n/Ling284, Natural language processing with deep learning January 2022–February 2022
- Open course: Introduction to reinforcement learning May 2021– August 2021
- Open course: Stanford CS224w, Graph neural networks May 2021– August 2021
- Open course: UCB CS285, Deep reinforcement learning May 2021– August 2021
- Open course: Next Step of Machine Learning January 2021– February 2021
- Open course: Geometric Deep Learning June 2022 – September2022
- Open course: Stanford CS330 Deep Multi-Task and Meta Learning September2022 – December 2022
- Open course: Categories for AI March 2023 – December 2023

SKILLS

Programming: C, C++, Python (Pytorch, Gym, Ray(RLlib), Triton, PsychoPy), CUDA, HTML, FPGA, JavaScript
Applications: Matlab, WordPress
Languages: English (TOFL BEST SCORE 2025: 113/120). Mandarin Chinese (First language)